STORAGE



Recent Advancements in OCP Telemetry for SSDs

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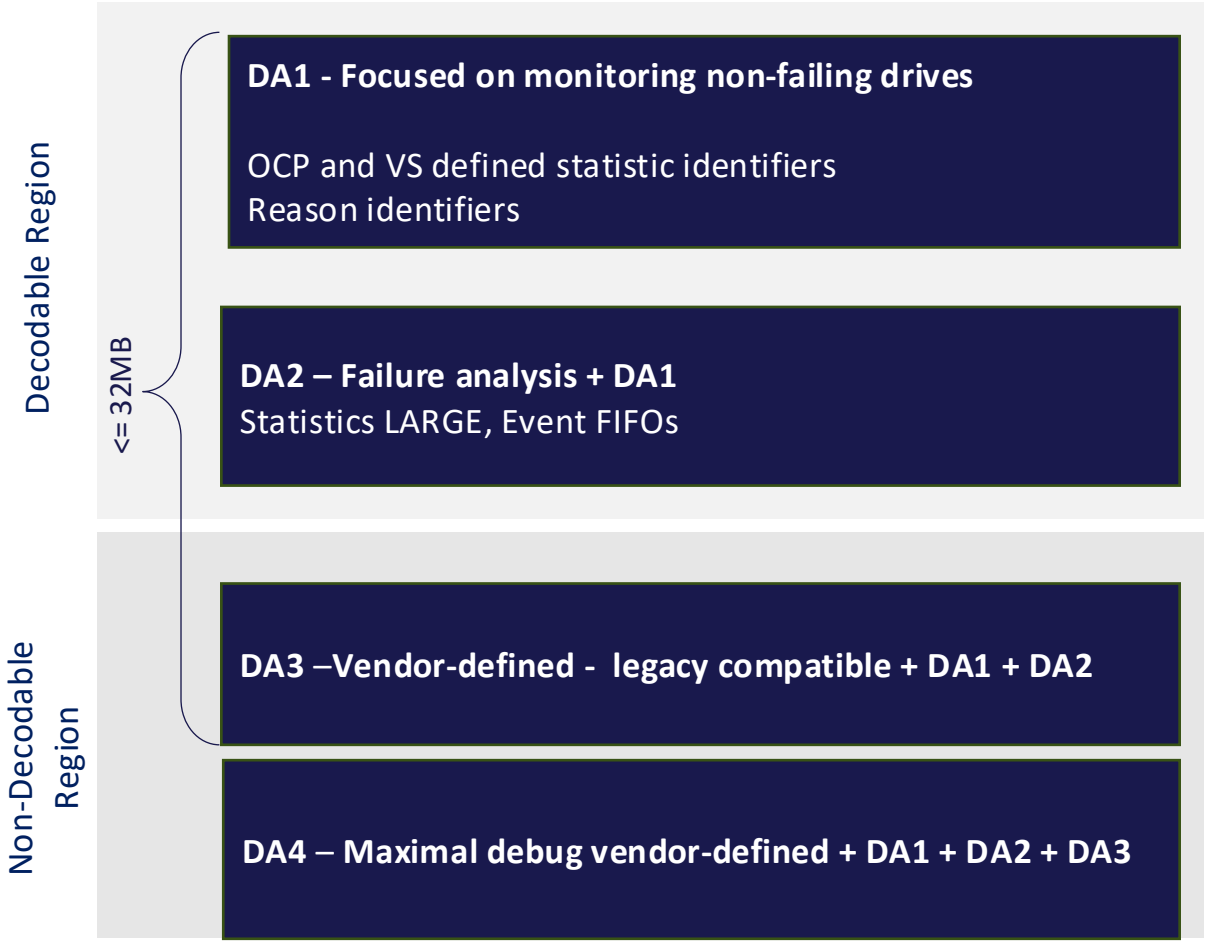
Dhankaran Ajravat *SSD Component Quality Engineer, Meta*





OCP DC NVMe SSD Telemetry Overview

- Built on NVMe Telemetry Log Pages



OCP Datacenter NVMe SSD Specification Version 2.7

Enables:

- Standardized Statistics and Events
- Human readable output for first pass debug
- Standardized debug tools
  - nvme-cli with OCP Plugin
- Customer and vendor validating the same tools

Results:

- Faster deployment of tools to the datacenter
- Confidence that customer data is not exposed
- Enables pro-active fleet monitoring

Proactive SSD Health Monitoring

CHALLENGE

- Millions of SSDs deployed, diverse supplier base.
- Failures detected only **AFTER** they happen **REACTIVE**
- At scale, lower %es have high impact → Unplanned replacements disrupt capacity, lower QoS.
- No fleet-wide degradation visibility.

SOLUTION (WIP)

- Fleet-wide analytics (SMART, ext., DA1/DA2, **LM** logs*).
- Track anomalistic behavior - variance from population means/P90..
- Degree of variance define **RISK**, drives response.
- Vendor collaboration improves detection algorithms and/or product (WIN-WIN)

**OCP Latency Monitoring deployed on fleet SSDs to detect tail latencies missed by host-level* checks.

PROCESS OVERVIEW

META

- Dashboard: Actively monitors/ notifies anomalies (variance from mean, Risk). **PROACTIVE**
- Share parametric, DA1/2 data with Micron.

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- Provides list of Drives at Risk, remediation action; algorithm/product improvement plan.

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- Trend analytics across relevant SMART/counters (WA, Media errors, E2E, multi parameter correlation).

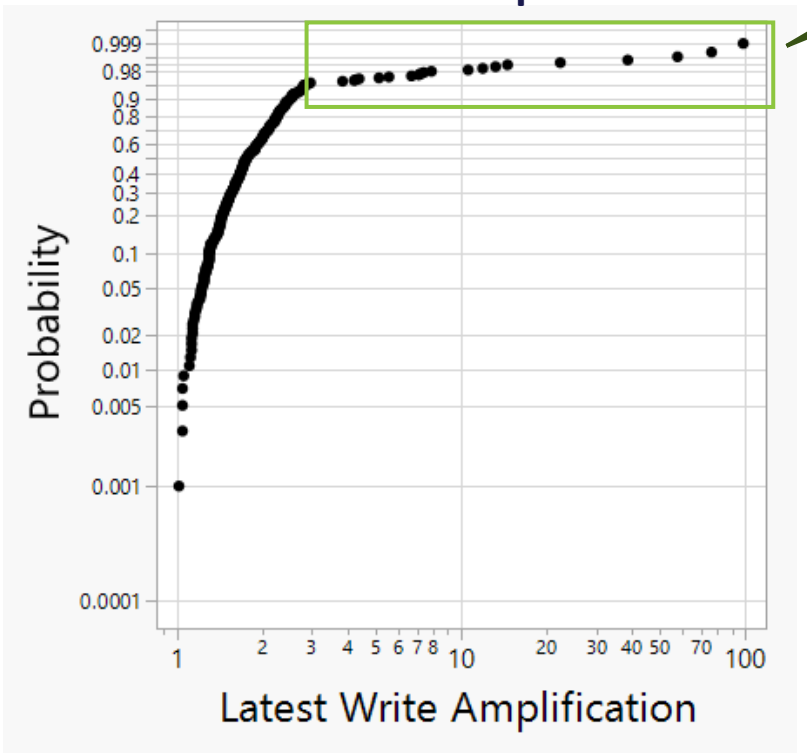
META

- Pre-emptive remediation (**unplanned** to **planned**)
- Improve dashboards' anomaly detection algorithm.

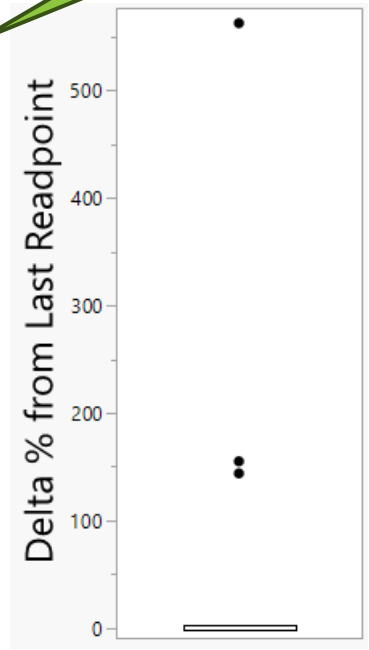
Examples

Representative data – not actual Micron SSDs

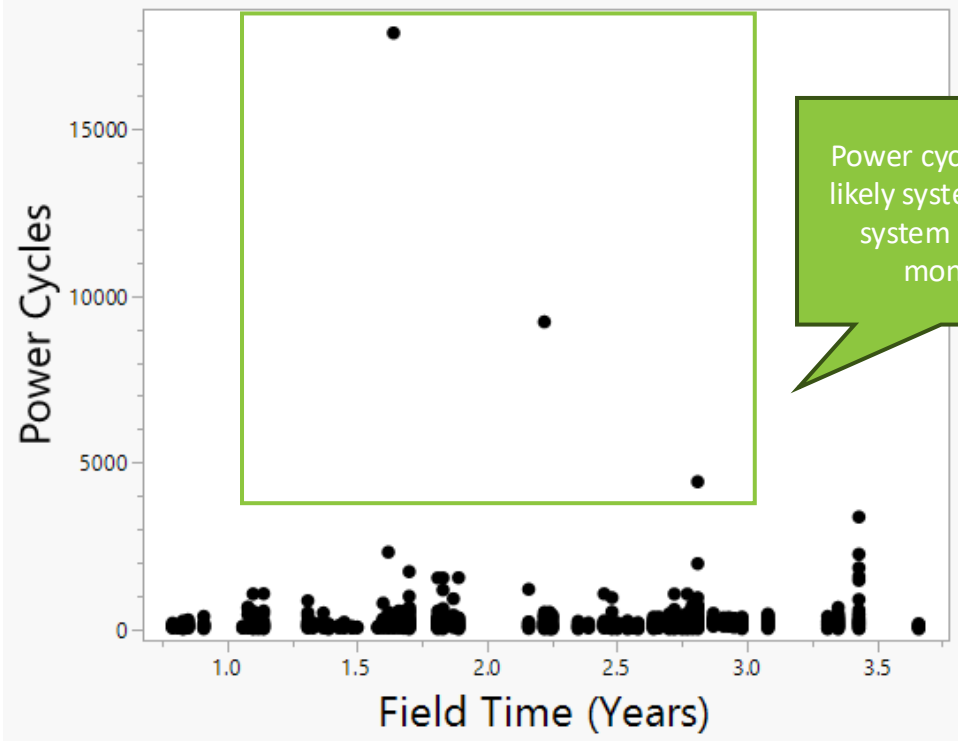
Write Amp



WAF outliers – could be drive or system caused



Power-Cycles

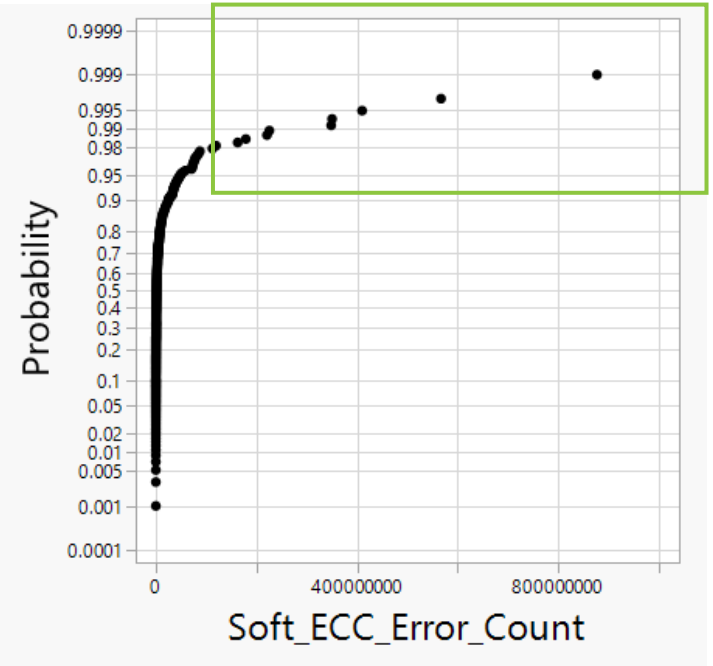


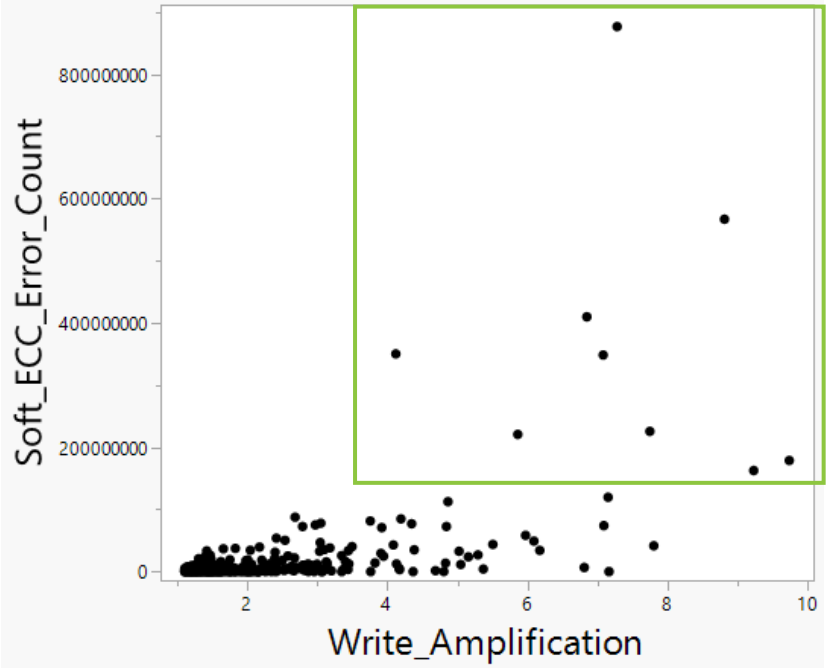
Power cycle outliers – likely system caused - system should be monitored

- Data is analyzed and recommendations for retirement /monitoring of both drives and systems is made
- Outliers identified by using a grouping of SMART attributes and statistics
- Medium risk drives and trends are analyzed

Representative data – not actual Micron SSDs

Examples (Trends)





- NAND reads not recoverable by first level ECC
- Slight trend of higher write amplification for drives with higher soft ECC error count

Credits: Jacob Brooksby (Micron)

Representative data – not actual Micron SSDs

Example Actions to Be Taken

Drives to Retire

- The following drives show some items of concern
- Recommendation is to retire
- Micron requests a subset of the drives be sent back to Micron for analysis

Serial Number	Reason	Recommended Action	Replacement Criticality	Return to Micron?	APPROX FIELD YEARS	write_amp	READ AMP	uc_err	media_errors	percent_used	available_spare	bad_block_count	soft_ecc_error_count	e2e_detected_error	refresh_count	REFRESH PER HOST GB WRITTEN	DELTA EC	APPROX AVG EC	xor_recovery_count	num_err_log_entries	critical_warning	vmbu_failed
		Retire	When possible	Yes																		
		Retire	When Possible	Yes																		
		Retire	Within a Month	Yes																		
		Retire	When Possible	Yes																		
		Retire	When Possible	Yes																		
		Retire	When Possible	No																		
		Retire	When Possible	No																		
		Retire	Within a Month	Yes																		
		Retire	Within a Month	Yes																		
		Up to Meta	N/A	No																		
		Up to Meta	N/A	No																		

OCP Latency Monitoring

- **Limitations:** Host-level checks miss extreme tail latencies (P99.999999+) due to their transient nature.
- **Challenge:** Intermittency makes debugging & data collection difficult for vendors.
- **Solution – Latency Monitoring**: Enables precise detection of tail latencies & stalls, improve visibility and SSD-level diagnostics.
- Thousands Of Drives Enabled SSDs with Latency Monitoring active.
- **Call to Action:**
  - Enable automation/analytics of LM/telemetry data from fleet.
  - Correlations between LM data and structured telemetry logs/human-readable logs.

FINAL TAKEAWAY & CALL TO ACTION

Driving Impact Through Collaboration

REACTIVE

↓

PROACTIVE

UNPLANNED

↓

PLANNED

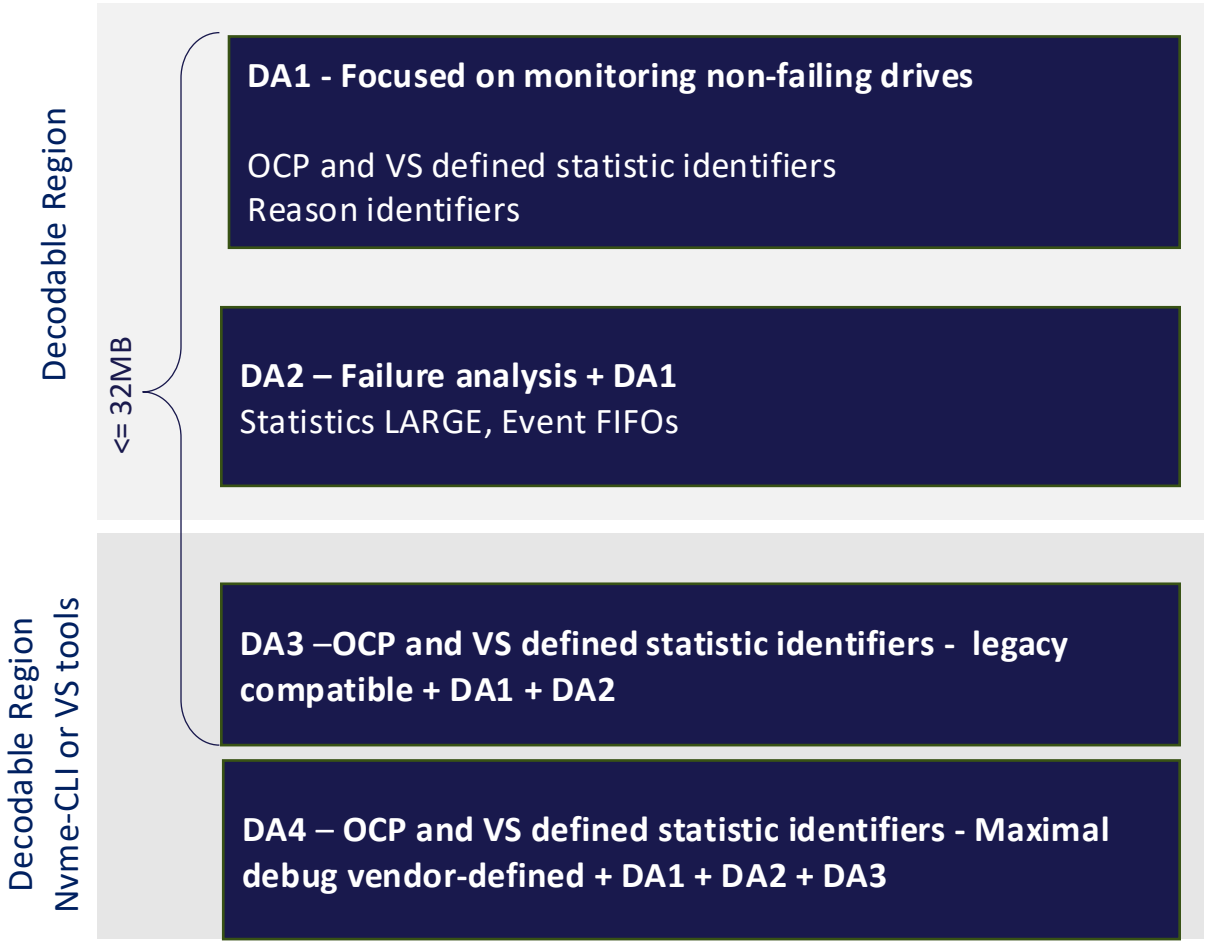
COLLABORATION ENABLES

- Improved Diagnostics
- Improves Fleet Reliability
- Better Future Products

Vendor partnership transforms maintenance from cost center to strategic advantage

Call To Action: implement OCP telemetry features.

Recent Advancements in OCP Telemetry



- OCP 2.8 will expand the NVMe-CLI/OCP-plugin human-readable region to include DA3 and DA4!
  - Drive can populate OCP-defined FIFOs and events in DA3 and DA4
- Drive can advertise:
  - DA3/DA4 – fully human-readable (via NVMe-CLI/OCP plugin)
  - DA3/DA4 – partially contains human-readable data and contains some vendor tool human-readable data
- Using the Telemetry Management feature host can select supported behaviors
- These improvements apply to latency monitor debug log as well!

- We anticipate that this capability will be released into nvme-CLI OCP plugin in early 2027.

Recent Advancements in OCP Telemetry

Challenge:

NVMe-CLI and OCP plugin are not represented on all operating systems

Solution:

- Micron has been actively collaborating with the open-source community to port NVMe-CLI and the OCP plugin to Windows.
- This will align the tool chain and drive debug consistency across platforms.

NVMe-CLI 3.0 for Windows is Released!

https://github.com/Micron-TPG-OSS/nvme-cli/releases/tag/v3.0

```
user@TESTSYSTEM C:\Users\user\Development\nvme-cli\.build-static-json>ver

   Microsoft Windows [Version 10.0.26100.32522]

user@TESTSYSTEM C:\Users\user\Development\nvme-cli\.build-static-json>.nvme ocp internal-log PHYSICALDRIVE2 --data-area=2 --output-format=json
   {
     "result":"\nMissing telemetry-log. Fetching from drive...\n"
   }
   {
     "result":"telemetry.bin generated. Proceeding with next steps.\n"
   }
   {
     "result":"Missing string-log. Fetching from drive...\n"
   }
   {
     "result":"string.bin generated. Proceeding with next steps.\n"
   }
   Extracting Telemetry Host Dump (Data Area 2)...
   ocp internal-log command completed.

user@TESTSYSTEM C:\Users\user\Development\nvme-cli\.build-static-json>
```

- Screenshot: OCP Telemetry Log pull and decoding with OCP Plugin with Windows nvme-cli build

Call to Action

- Deploy industry standard telemetry. Not sure how to capture and decode telemetry data?
  - See Micron tech note
- Have ideas on improving SSD Telemetry? Sign the OCP DC NVMe SSD Feedback CLA and get involved!

Micron Telemetry Tech Note

Thank You!

